

Phaneendra Gavara

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SUMMARY

AI/ML Engineer with hands-on experience spanning production agentic AI systems, RAG/LLM evaluation infrastructure, and applied machine learning — including a solo-built, AWS-deployed agent platform, an automated CI/CD RAG evaluation pipeline, and PyTorch deep learning models. M.S. Data Science graduate of Arizona State University (GPA 3.90/4.0).

EDUCATION

Arizona State University <i>M.S. Data Science (Computing & Decision Analytics Concentration)</i> — GPA: 3.90/4.0 • <i>Relevant Coursework:</i> Statistical Machine Learning, Data Mining, Data Processing at Scale, Computing & Data-Driven Optimization, Data Science Capstone	Tempe, AZ Aug 2024 – May 2026
IIIT Bhubaneswar <i>B.Tech in Computer Science and Engineering</i> — GPA: 7.14/10	Bhubaneswar, India Dec 2020 – May 2024

TECHNICAL SKILLS

Agentic AI & LLMs: LangGraph, LangChain, Claude API, GPT-4o, Model Context Protocol (MCP), Multi-Agent Orchestration, RAG, Prompt Engineering
ML & Deep Learning: PyTorch, scikit-learn, CNNs, DistilBERT, SARIMAX, OR-Tools, Pandas, NumPy
AI Evaluation & MLOps: RAGAS, Automated Evaluation Suites, Quality Gates, Hallucination Rate Scoring, Docker, Kubernetes, CI/CD (GitHub Actions)
Backend, Cloud & Languages: FastAPI, PostgreSQL, Pinecone, AWS (EKS, Lambda, S3), Azure OpenAI, Python, SQL, TypeScript

PROJECTS

AgentForge — Agent-Native Platform <i>Python, FastAPI, React, TypeScript, AWS (EKS, Lambda, S3), Docker, PostgreSQL</i> - Solo-architected and built a production agent-native platform (“Vercel for AI agents”) to build, deploy, evaluate, and govern autonomous AI agents, spanning 5 microservices and ~16,800 lines of code - Deployed the full platform live on AWS EKS behind HTTPS and a WAF, then automated a complete teardown to \$0 idle cost, demonstrating end-to-end cloud lifecycle ownership - Wrote 160+ automated tests across services to validate agent evaluation and governance workflows before deployment	2026
ASU LLM Evaluation Pipeline <i>Python, LangChain, RAG, RAGAS, GPT-4o, Pinecone, GitHub Actions, React</i> - Built an automated CI/CD evaluation pipeline that scores a university RAG chatbot across 100 test questions on every code push, enforcing 6 automated quality gates before deploy - Cut LLM-judge evaluation cost by 25x and sped up result aggregation by 8x through batched scoring and caching - Integrated RAGAS-based automated scoring with a Pinecone vector store and a GPT-4o judge to quantify RAG quality without manual review	2026
InsightHub — AI-Powered Business Analytics Platform <i>Python, FastAPI, Azure SQL, Azure Blob Storage, Azure OpenAI (GPT-4o), Azure AI Search, React</i> - Designed and deployed a full-stack Azure analytics platform ingesting 230,000+ rows through a star-schema ETL pipeline into Azure SQL - Built a FastAPI backend serving business insights to a React dashboard, fully deployed and live in production - Implemented a GPT-4o-powered RAG search engine over the analytics data so users can query business metrics in natural language	2026
TravelIQ — AI Travel Itinerary Generator <i>Python, Flask, OR-Tools, scikit-learn, SARIMAX, GPT-4o</i> - Built an AI travel planner that generates a complete, feasibility-checked itinerary in under 90 seconds, combining ML recommendations, image generation, and route optimization - Modeled demand and pricing trends with SARIMAX time-series forecasting and scikit-learn to inform itinerary recommendations - Solved multi-stop routing as a Vehicle Routing Problem with Time Windows (VRPTW) using OR-Tools to guarantee feasible schedules	2026

Republic of Bean — Agentic AI Ordering System

LangGraph, Claude API, FastAPI, PostgreSQL, Model Context Protocol (MCP)

- Built a multi-agent conversational ordering system using LangGraph with Claude as the orchestrating agent across specialized sub-agents
- Exposed PostgreSQL-backed internal ordering APIs to the LLM through the Model Context Protocol (MCP), decoupling agent reasoning from application logic
- Designed the FastAPI backend to handle concurrent multi-turn conversations while maintaining consistent order state

2025

Anomaly Detection in Crowds

Python, PyTorch, CNNs, OpenCV, Deep Learning

- Built a CNN-based crowd safety system in PyTorch that learns normal crowd behavior from video and flags anomalies in real time
- Generated frame-level heatmaps to visualize and localize anomalous activity for rapid human review
- Processed video streams with OpenCV for frame extraction and preprocessing ahead of model inference

2025

Heart Disease Detection

Python, scikit-learn, Pandas, Matplotlib, Seaborn, SHAP

- Built a supervised ML classifier achieving 85% accuracy on the Cleveland Heart Disease dataset
- Benchmarked multiple model architectures and used SHAP feature analysis to explain individual predictions and surface the strongest risk factors

2024

EXPERIENCE

Visiting Researcher

Arizona State University

- Designing, developing, and implementing AI-based approaches and solutions for blockchain technologies
- Enhancing and analyzing machine learning models, algorithms, and intelligent blockchain applications
- Researching and implementing AI techniques to improve blockchain security, scalability, and efficiency

Aug 2026 – Present
Tempe, AZ

Teaching Assistant & Grader

Arizona State University, Ira A. Fulton Schools of Engineering (CSE 230 – Computer Architecture)

- Graded student submissions and gave feedback across cache memory calculations, AMAT, MIPS multi-cycle control signals, and pipeline hazards using structured, leniency-calibrated rubrics
- Held office hours and managed course communications (Canvas announcements, student Q&A) for a Computer Architecture course

Aug 2025 – May 2026
Tempe, AZ

CERTIFICATIONS

- AWS Certified Solutions Architect – Associate (Jul 2026)
- AWS Certified AI Practitioner (Jun 2026)
- AI Fluency Framework & Foundations – Anthropic (May 2026)
- Claude Code 101 – Anthropic (May 2026)
- Claude 101 – Anthropic (May 2026)
- Introduction to Claude Cowork – Anthropic (May 2026)

LEADERSHIP & AWARDS

Admin Assistant & Executive Member

Help N Hope Charitable Society (NGO)

- Co-founded the organization and was appointed Executive Member (2024) for a sustained commitment to making quality education accessible to underprivileged children
- 3rd Place, Social Bias Hackathon – Ethical Spectacle Research x Arizona State University (Oct 2024)

Jan 2021 – Present
Andhra Pradesh, India